EXPERIMENT NO 8

Create an infographic in Figma explaining the concept of Public Land Mobile Network (PLMN).

Aim:

To create an infographic in Figma explaining the concept of Public Land Mobile Network (PLMN).

Procedure:

1. Open Figma
2. Create a new file
3. Select the Frames
4. Fill in the content that is required for presentation
5. Design Visual Elements
6. Make it Interactive
7. Add Annotations and Explanations
8. Incorporate Multimedia
9. Storyboard Animation
10. Review and edit the Prototype
11. Save and Share

Components and Their Functions

1. Mobile Station: user device.
2. Radio access network: connects device to operator.
3. Core network: routes services and manages mobility.
4. Services: voice, SMS, data and roaming.
5. MCC/MNC: codes identifying country and mobile network.

FIGMA LINK :

https://www.figma.com/design/Xdb7ymUwwBCfUBK8X6D4Qq/Untitled?node-id=4-2&p=f&t=vO0Naw90BAKGJv0b-0

OUTPUT :


Experiment 8 — Public Land Mobile Network infographic
08
MOBILE COMPUTING / EXPERIMENT 8
Public Land Mobile Network (PLMN)
A PLMN is a mobile communication network operated by a service provider for public mobile services.
Public mobile services
Operator-managed infrastructure
Mobility across networks
01 Main components
Classic GSM reference architecture
These seven components describe the classic GSM model. Later cellular generations use different core-network functions and names.
MS
Mobile Station
The user's handset and SIM; accesses the network over the radio interface.
Radio
BTS
Base Transceiver Station
Transmits and receives radio signals to serve mobile stations in a cell.
Control
BSC
Base Station Controller
Controls multiple BTSs, radio resources and handovers within its area.
Switching
MSC
Mobile Switching Center
Switches calls and coordinates mobility and connections to other networks.
Subscriber & location queries
AuC
Authentication Center
Holds secret keys and generates authentication and ciphering data for subscriber security.
Authentication data
HLR
Home Location Register
Stores permanent subscriber profiles and the current serving network location.
VLR
Visitor Location Register
Keeps temporary subscriber and location data for users in an MSC's service area.
Simplified functional links: arrows show the service path or query direction; signaling responses travel back over the same links.
02 Working process
Simplified service path
Mobile phone
Initiates a call or service
Base Station
Provides the radio link
Switching Center
Routes the connection
Destination Network
Delivers to the recipient
In GSM, the BSC manages the radio path between the BTS and MSC. The MSC uses subscriber records to help establish and route the service.
03 Main services
Voice calls
Connect subscribers through call switching.
SMS
Exchange short text messages.
Mobile data
Access data services over a mobile connection.
Roaming
Use services on a visited operator's network.
Subscriber authentication
Verify a subscriber using SIM-based credentials.
04 Network identity
Which operator's network?
PLMN identity commonly includes a Mobile Country Code (MCC) and a Mobile Network Code (MNC).
Together, they identify the mobile network, not an individual subscriber. The MNC is assigned within the MCC.
MCC
3 digits
Mobile Country Code
+
MNC
2 or 3 digits
Mobile Network Code
Architecture note: packet data in GSM commonly uses GPRS/EDGE with additional nodes such as SGSN and GGSN, not shown in this classic GSM diagram.

RESULT :

Hence created an infographic in Figma explaining the concept of Public Land Mobile Network (PLMN).